

SNR RANGE

Typical LoRa SNR limits

Spreading Factor	Approx. Minimum Usable SNR
SF7	~-7.5 dB
SF8	~-10 dB
SF9	~-12.5 dB
SF10	~-15 dB
SF11	~-17.5 dB
SF12	~-20 dB

For your SF7–SF12 configuration

- **SNR > 0 dB:** Excellent signal
- **SNR -5 to -10 dB:** Good/usable depending on SF
- **SNR -10 to -15 dB:** Weak
- **SNR -15 to -20 dB:** Very weak / near the limit
- **SNR < -20 dB:** Generally consider **NO SIGNAL / unreliable**

Important for your self-healing LoRa P2P system

Don't use a single fixed threshold for every SF. For example:

- At **SF7**, SNR -15 dB is effectively unusable.
- At **SF12**, SNR -15 dB can still be a valid and relatively decodable signal.

A practical conservative rule could be:

```
SNR < -20 dB → NO SIGNAL
SNR -20 to -15 dB → Critical
SNR -15 to -10 dB → Weak
SNR -10 to 0 dB → Good
SNR > 0 dB → Excellent
```

For your **maximum-range configuration using SF7–SF12**, I would recommend treating **around –18 to –20 dB as the worst-case operational SNR**, with **–20 dB as a practical failure threshold**.